Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/25/2022
PI Name:	McDonald, Neil Ph.D.		
Project Title:	Individualized, Noninvasive Speech Indicators for Tracking Elevations in Stress (INSITES)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	neil@quasarusa.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	858-373-0231
Organization Name:	Quantum Applied Science And Research, Inc.		
PI Address 1:	16744 West Bernardo Drive		
PI Address 2:			
PI Web Page:			
City:	San Diego	**State:**	CA
Zip Code:	92127-1904	**Congressional District:**	52
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	07/06/2020	**End Date:**	07/05/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC20C0118		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The complexity and round-the-clock nature of NASA operations in low Earth orbit (LEO) and future cis-lunar deep space missions, along with isolation in the extremely hostile environment of space, can induce levels of acute and chronic stress that could compromise astronaut performance, leading to errors that could affect science payloads, crew safety, and mission success. For the exploration of space, therefore, a method is needed to assess operator state, quickly and reliably detect stress, and provide objective feedback to the individual, crew, and ground support, in order to mitigate adverse events and mishaps. Due to the unique challenges of NASA missions, Quantum Applied Science and Research (QUASAR) proposes to develop a system to identify Individualized, Noninvasive Speech Indicators for Tracking Elevations in Stress (INSITES). The overall goal of this INSITES project is to develop an unobtrusive, objective, and reliable detector of stress that measures changes in speech and vocalizations using equipment that would be present or used (microphones, communications systems, computers) during operations, thus not requiring additional sensors or dedicated processing hardware. QUASAR and the Florida Institute for Human and Machine Cognition		

	(IHMC) will build a database of audio stress recordings acquired under laboratory conditions in order to construct normative models of stress, using vocal stress-related features identified in Phase I. A methodology for recalibrating normative models to individuals using minimal additional training data in order to optimize model performance will be developed. QUASAR will also prepare and validate an INSITES prototype that will provide a real-time visual output describing an individual stress level. The prototype will be based upon an app that can be readily installed on a mobile device or implemented in NASA spacecraft and habitats to detect changes in stress acutely and over time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Non-NASA Applications: Multiple markets across both military and civilian mission critical environments where personnel operate and communicate in stressful environments. In particular, this technology could extend to military and commercial pilots, air traffic control operators, security or first response teams, as well as elite performance teams where audio communication is enabled by wearable headsets.
Task Progress:	New project for FY2020. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)